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324711 (24)

BE (7th Semester)

Examination, April-May, 2013

Branch : Electrical

HIGH VOLTAGE ENGINEERING

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt part (a) and any two parts from (b), (c) and (d) of each question.

Q. 1. (a) Define Townsend's first and second ionization coefficients. 2

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(b) Explain various ionization processes taking place in the gases. 7

(c) Explain the mechanism of development of anode and cathode streamers and explain how these lead to breakdown. 7

(d) Find the total corona loss in a 132 KV, 50 Hz, 3 phase transmission line, 200 km long consisting of three 2 cm diameter stranded copper conductors spaced in 3.5 m delta arrangement for an over voltage of 2 per unit. The atmospheric temperature is 26°C and barometric pressure is 740 mm Hg The roughnes coefficient is 0.8 7

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Q. 2. (a) Define intrinsic strength of a solid insulating material. 2

(b) What are "Treeing and Tracking" ? Explain clearly the two processes in solid dielectrics. 7

(c) State various processes which lead to formation of bubbles in liquid dielectrics and explain clearly cavity breakdown mechanism in liquid dielectrics. 7

(d) Explain the breakdown due to voids in a solid dielectrics. A solid insulating block of thickness 1.5 c.m. and having a dielectric constant of 4.5 is subjected to a 50 Hz a.c.

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voltage. The specimen contains an air void of thickness 1 mm. Considering the breakdown strength of air as 30 KV/cm (peak), find the maximum voltage which can be applied across the specimen without any internal discharge. 7

Q. 3. (a) Define ripple voltage of a rectifier circuit. 2

(b) Explain with neat diagram Cockroft-Walton multiplier circuit. Explain clearly its operation when the circuit is (i) unloaded (ii) loaded. 7

(c) Explain a three stage cascaded transformer for the generation of high A.C. and determine the expression for short circuit impedance of the transformer. 7

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(d) Explain with a neat diagram triggering and synchronisation of the impulse generator with CRO. 7

Q. 4. (a) What are delay cables ? 2

(b) Draw a neat schematic diagram of a generating voltmeter and explain its principle of operation. Discuss its application and limitations. 7

(c) Explain the construction of sphere gap for measurement of AC high voltages and discuss the procedure for the same. 7

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(d) Discuss the basic working principle of an electrostatic voltmeter in brief. An absolute electrostatic voltmeter has a movable circular plate 8 cms in diameter. If the distance between the plates during a measurement is 4 mm, determine the potential difference when the force of attraction is 0.2 gm wt. 7

Q. 5. (a) Describe the function of automatic guard potential regulator used in high voltage Schering bridge.

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(b) Explain briefly impulse testing of power transformer. 7

(c) List out various tests to be carried out on cable and give a brief account of each test. 7

(d) Explain the operation of high voltage schering bridge when the test specimen is grounded. 7

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